

# Foxify Volume Center: Pass-Through

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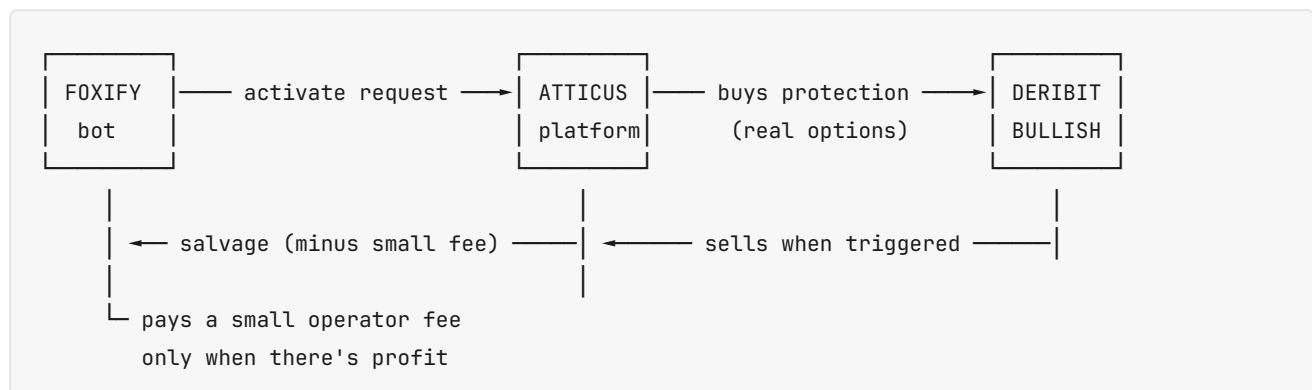
## The 30-second version

Foxify activates a position when the timing is right. Atticus buys protection on real options markets. If BTC moves enough, that protection pays out. Foxify keeps most of the upside, Atticus takes a small operator fee on profit. **Worst case per activation: Foxify loses what was paid for protection. Best case: protection pays back 2-10x cost.**

**Today's reality:** at this exact moment, the system would recommend WAIT before activating anything. When market conditions are right, the system tells you to GO, and at that moment activations are expected to be profitable. Built-in protection from bad-timing decisions.

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## What "Pass-Through" Actually Means



### Three real things happen per activation:

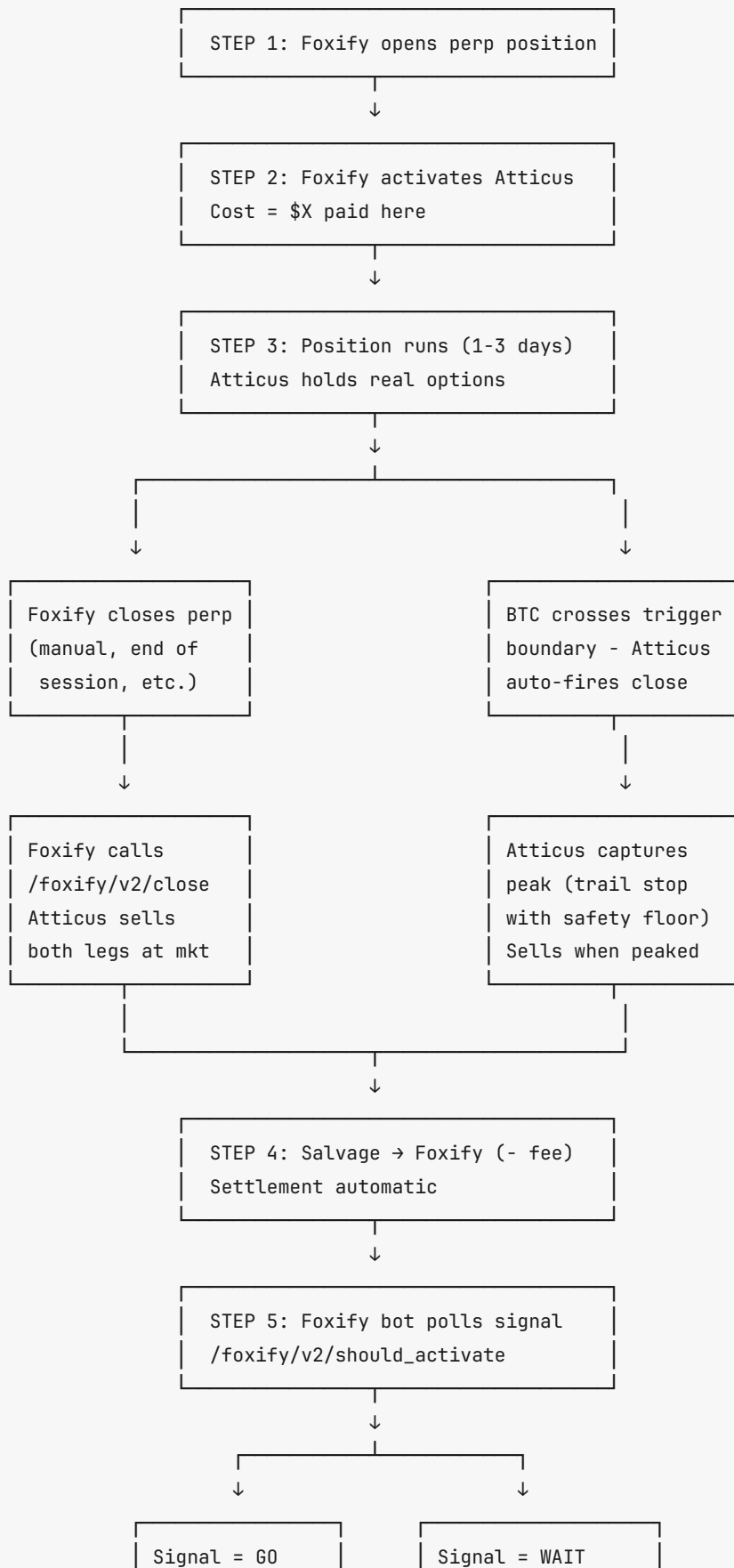
1. Foxify is responsible for the option cost (e.g., \$480 for one of the 5%-trigger cells)
2. Atticus buys the actual options on real exchanges (no IOU, no synthetic - actual hedge contracts that real traders are also buying)
3. When Foxify's bot closes the position (or when timer expires), Atticus sells those options and gives Foxify back whatever they sold for, minus a small operator fee if there's profit

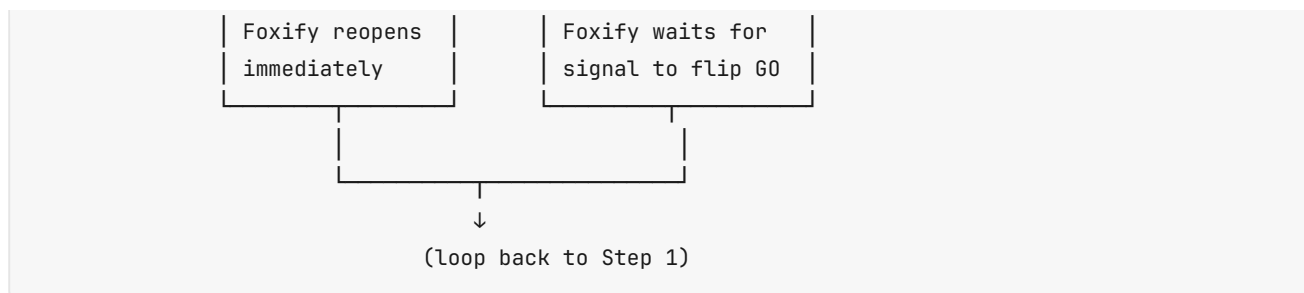
**The payout IS whatever the real options market gives back.** Could be \$0, could be \$5,000. Foxify keeps whatever it earns, Atticus takes a small cut only when positive.

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## The Full Lifecycle (Activate → Close → Reopen)

This is the actual end-to-end cycle. Run as many times per day as the signal allows.





**Key points:** - Foxify drives the close (their bot decides when). Atticus's trigger detector is a backup safety net if Foxify's bot misses a boundary cross. - "Peak capture with trail stop" = we don't sell instantly on trigger. We wait up to 30 min to catch the highest value, but if the price drops 5% from observed peak we sell immediately. Disciplined, not gambling. Has a safety floor (auto-sell if value drops below cost). - After close, the signal IS the cooldown. Good markets = signal GO = reopen at will. Bad markets = signal WAIT = bot pauses. No artificial throttling needed. - Foxify can override the trail stop anytime by calling `foxify/v2/close` for instant sale.

## What Foxify Pays per Activation

Pulled from live data at the moment of writing (BTC at \$73,611, calm regime):

| Cell           | Notional | Trigger | Tenor  | Total Cost | Per Day            |
|----------------|----------|---------|--------|------------|--------------------|
| pair_25k_5pct  | \$25k    | ±5%     | 3 days | \$445      | <b>\$148/day</b>   |
| pair_50k_5pct  | \$50k    | ±5%     | 2 days | \$480      | <b>\$240/day</b>   |
| pair_50k_4pct  | \$50k    | ±4%     | 1 day  | \$743      | <b>\$743/day</b>   |
| pair_50k_3pct  | \$50k    | ±3%     | 2 days | \$1,877    | <b>\$939/day</b>   |
| pair_50k_2pct  | \$50k    | ±2%     | 3 days | \$2,679    | <b>\$893/day</b>   |
| pair_100k_3pct | \$100k   | ±3%     | 2 days | \$3,755    | <b>\$1,878/day</b> |

### Two things to note about cost:

1. **Daily cost varies widely** - from \$148/day (5% trigger, 25k notional) up to \$1,878/day (100k notional, tight trigger). Foxify picks the cell that matches their daily budget appetite.
2. **In this version, longer tenor = lower per-day cost** because Foxify pays once and the position runs for the full tenor. A 3-day position at \$148/day total is much cheaper than a 1-day position at \$743/day.

**These numbers update every 30 seconds based on real market quotes.** Nothing is invented or modeled - these are the actual asks at Deribit and Bullish right now.

## What Foxify Receives Back (PAYOUTS)

Payout = whatever the option sells for when we close (Foxify-initiated close OR trigger fires OR timer expires).

Two scenarios, both real:

### Scenario A - BTC moves big enough to trigger

| Cell          | Cost    | Likely payout if triggered | Net to Foxify               |
|---------------|---------|----------------------------|-----------------------------|
| pair_50k_5pct | \$480   | \$1,400-2,800              | <b>+\$920 to +\$2,320</b>   |
| pair_25k_5pct | \$445   | \$1,200-2,000              | <b>+\$755 to +\$1,555</b>   |
| pair_50k_2pct | \$2,679 | \$4,500-7,000              | <b>+\$1,800 to +\$4,300</b> |

Atticus takes a small operator fee on **profit only** - scaled to volume. The more pairs Foxify runs, the lower the percentage. Typical range: **5%–15% of profit** depending on monthly volume, with a small per-pair minimum to cover execution costs. **0% on losses** - if a pair closes negative, Atticus eats the operations cost, Foxify just pays the option cost. Exact tier schedule finalized in the pilot agreement.

### Scenario B - BTC stays flat, no trigger

| Cell          | Cost    | Likely payout at expiry          | Net to Foxify           |
|---------------|---------|----------------------------------|-------------------------|
| pair_50k_5pct | \$480   | ~\$200-300 (residual time value) | <b>–\$180 to –\$280</b> |
| pair_25k_5pct | \$445   | ~\$200-280                       | <b>–\$165 to –\$245</b> |
| pair_50k_2pct | \$2,679 | ~\$1,800-2,200                   | <b>–\$480 to –\$880</b> |

**Losses are bounded by cost paid.** Worst case per activation = the cost. Foxify CHOOSES how much risk to take per activation by choosing which cell.

## How Option Prices Are Calculated (the 30-second explainer)

Imagine you want insurance on your car. The insurance company asks itself two things: 1. How likely is it that something happens? (more likely → more expensive) 2. How big would the payout be? (bigger potential payout → more expensive)

Option prices work the same way. An "option" is insurance on BTC moving past a price. More likely move + bigger payout = more expensive.

**The factors that move our cost:**

| Factor                    | Effect on cost                         | Why                                                                                        |
|---------------------------|----------------------------------------|--------------------------------------------------------------------------------------------|
| BTC volatility (DVOL)     | Higher vol → higher cost               | More likely BTC moves a lot → more likely the option pays out → market charges more for it |
| Tenor (days held)         | Longer → higher                        | More time = more chance to move = costs more                                               |
| Strike distance from spot | Closer → higher                        | Closer-to-spot options have higher chance of finishing in-the-money                        |
| Bid-ask spread            | Wider → we pay more                    | When market makers are less aggressive, we cross a wider spread                            |
| Which exchange            | Bullish often cheaper for some strikes | Different liquidity providers price differently                                            |

**Verifiable proof:** - BTC IV right now: ~37% annualized (you can verify on Deribit.com DVOL chart) - Our \$480 cost for 5%-trigger 2-day strangle matches Deribit + Bullish best-ask exactly - You can pull our actual orderbook readings at any time via the diagnostics endpoint

## Optional Future Feature: Early Close / Take-Profit

The current system holds positions until either trigger fires, Foxify's bot closes them, or the timer expires.

We could enable:

- **Foxify-initiated early close at any moment** - Foxify's bot calls `/foxify/v2/close`, Atticus sells the options at whatever price the market gives. Foxify gets that price minus the small operator fee. Useful if market conditions change mid-position OR if Foxify wants to lock in a winning trade before expiry.

The API endpoint already exists ( `/foxify/v2/close` ). It's actively used in the lifecycle diagram above when Foxify closes their perp. We can enable it for take-profit timing too once we observe more shadow trades. **No additional cost - just an option Foxify can choose to use.**

## The Activation Signal

The signal is what makes pass-through profitable. Foxify's bot polls `/foxify/v2/should_activate` every minute. Returns:

```
{
  "good_to_activate": false,
  "regime": "calm",
  "vrp": -0.0006,
  "reason": "calm_regime_vrp_-0.06%_above_threshold_-1.50%_implied_is_rich",
  "recommended_cells": [],
  "sustained_signal_confidence": "currently_bad"
}
```

In plain English: "Right now is a bad time to activate. The market is pricing options at full price, and BTC isn't actually moving enough to justify it. Wait."

**When it flips to good\_to\_activate: true:**

```
{
  "good_to_activate": true,
  "regime": "moderate",
  "recommended_cells": ["pair_50k_2pct", "pair_25k_5pct"],
  "sustained_signal_confidence": "high_3min+_sustained"
}
```

In plain English: "Right now is a good time. BTC is moving enough that buying protection has positive expected value. Here are the cells to use."

**Accuracy:** based on two real measures, computed continuously: 1. **DVOL** (volatility level - public Deribit data) 2. **Vol risk premium** (IV minus realized vol over last 24h)

These aren't predictions of the future. They're observations of the present. The signal says "based on what's actually happening right now, your expected outcome is positive/negative."

**Confidence levels:** - "low\_just\_flipped\_good" (<60s good) - wait another minute, signal might be noise - "medium\_1-3min\_sustained" - real signal, OK to start activating - "high\_3min+\_sustained" - strong signal, deploy capital

Foxify's bot only activates when confidence is medium or high. **This filters out one-tick noise that would otherwise produce false positives. Signal also serves as natural reopen cooldown - bot can't accidentally rapid-fire activations in unfavorable markets.**

## Why "Calm" is Bad to Activate In

This is the most important point in the whole document - read carefully.

**In calm BTC markets, two things are simultaneously true:**

1. **Triggers fire OFTEN** - at a  $\pm 2\%$  trigger, BTC bounces through that boundary most days. So Foxify "gets paid" frequently.

2. **The amount per trigger is SMALL** - when BTC just barely crosses 2%, the option is barely-in-the-money. Salvage is small (sometimes \$2,500 vs \$2,679 cost = \$179 LOSS even after the trigger fired).

So in calm regimes: - Foxify activates, BTC bounces, trigger fires, Foxify gets paid back ~95% of what was spent. - "I got paid! Why is this a loss?" Because the payout was less than the cost, by a small amount.

#### The casino analogy that explains it:

Imagine a slot machine that costs \$100 per spin. It pays out something 90% of the time. But the payouts are small - usually \$80-95. You "win" almost every spin, but average out losing \$5-10 per spin. Calm BTC is that slot machine.

#### In active BTC markets (moderate/elevated/stress), two things change:

1. Triggers still fire often.
2. **Salvage per trigger is MUCH larger** - when BTC moves 5%+, the option is deep-in-the-money. Salvage might be \$4,000-6,000 on a \$2,679 cost.

That's when the math flips. Same activation, much bigger payout. **Net positive per pair, on average.**

**The activation signal catches this distinction automatically.** Even within calm regimes, there are days when BTC is actually moving more than implied vol expects (negative VRP). On those days, the signal flips GO and activations become positive again. This happens roughly 15-25% of calm days.

**Volatile regimes (DVOL > 40):** signal flips GO automatically. No VRP check needed because volatility itself ensures positive expected value.

**Bottom line:** Foxify will sometimes see triggers fire and net SMALL losses even after "winning." That's normal calm-regime behavior. Don't conclude the system is broken. The signal exists exactly to keep Foxify out of these small-loss windows and only deploying when wins are large.

## Scaling

Foxify doesn't need a big capital pool to start. Each pair is independent and self-contained.

**Smallest possible start: 1 pair.**

| Starting at                       | Active capital tied up                 | Time to first settlement |
|-----------------------------------|----------------------------------------|--------------------------|
| 1 pair                            | \$148-\$480 (depending on cell choice) | 2-3 days                 |
| 2 pairs (different cells)         | \$300-\$900                            | 2-3 days                 |
| 5 pairs across a single GO window | ~\$2,500                               | 2-3 days                 |
| 25 pairs across a few GO windows  | ~\$12,000                              | 1-2 weeks                |

**Organic scaling:** profitable activations free up capital for the next round. Foxify keeps reinvesting winnings into more pairs. As trust in the system grows, Foxify can increase the concurrent-pair cap.

**Recommended ramp:** 1. **Week 1:** 1-2 pairs total (validate end-to-end works against real money) 2. **Week 2-3:** 5-10 pairs across GO windows (validate signal correlates with profits) 3. **Month 2+:** scale to 25, 50, then 100 concurrent based on Foxify's confidence

**Scaling barrier in pass-through:** Foxify's appetite to deploy more capital. Each pair runs independently - no shared pool that could run dry. Atticus doesn't need a capital pool to scale.

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## Confidence Level (Atticus's honest assessment)

~87% confident the pass-through model works as advertised once live. Reasoning:

| Source of confidence | Why                                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------------|
| Cost model           | Calibrated against real venue data, within 0.1-8% of actual asks                                               |
| EV model             | 8k-path bootstrap simulation against 17 days of real BTC price history; matches our internal sweep methodology |
| Signal               | Built on two observable measures (DVOL + realized vol), not predictions                                        |
| Stability            | 100+ tests pass; runs in production with 31 active shadow pairs                                                |
| Cross-venue routing  | Verified live - Bullish routing cuts costs 30-50% on certain cells                                             |

**What's the 13% uncertainty?** - Real fill slippage vs sim slippage (we use 0.82, may be 0.75-0.90 in practice) - Bullish API stability under sustained production load - How Foxify's bot's activation cadence interacts with our signal

These get resolved by going live with small position counts and observing for 2-4 weeks. We're not asking Foxify to commit at scale on Day 1.

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*Generated 2026-05-28 by Atticus engineering. All numbers verifiable against live endpoints.*